

# TO YIN YU

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## SUMMARY

Entry-level full-stack software engineer (B.S. CS, UMD) shipping applied AI products end-to-end — realtime voice AI on OpenAI's Realtime API over WebRTC, FastAPI backends with Postgres + Redis, and production deployments across Vercel, Render, Neon, Upstash, and Netlify. Open to junior SWE roles.

## TECHNICAL SKILLS

**Languages:** TypeScript, Python, JavaScript, SQL

**Web & Backend:** Next.js, React, FastAPI, Tailwind CSS, PostgreSQL, Redis

**Applied AI & Realtime:** OpenAI Realtime API, WebRTC, ElevenLabs, LLM tool use, TensorFlow, Keras

**Infra & Deploys:** Docker, Git, Linux/Unix, Vercel, Render, Neon, Upstash, Netlify

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## PROJECT EXPERIENCE · Independent Software Projects, 2024 – Present

**SoloYap — Realtime Voice English Tutor** | Next.js, React 19, TypeScript, OpenAI Realtime API, WebRTC, Tailwind CSS, PWA

[soloyap.com](https://soloyap.com) · 2025

- Built a realtime voice English-practice app on OpenAI's Realtime API over WebRTC: learners speak through 13 CEFR-graded scenarios (A2–C1) or a free-chat mode and get instant grammar corrections and better-phrasing suggestions after every turn.
- Shipped as an installable PWA with optional Mandarin/Cantonese hints; cost-protected with per-IP rate limits, session caps, and an email-allowlist flow.

**SoloMock — AI Voice Mock Interviewer** | Next.js, TypeScript, OpenAI Realtime API, WebRTC, Monaco, Tailwind CSS, Vercel

[solomock.com](https://solomock.com) · 2025

- Built a realtime voice mock-interview app: the candidate speaks through coding problems while the AI probes complexity, gives Socratic hints, and reads code via debounced Monaco-editor snapshots streamed over a WebRTC data channel (9 languages).
- Authored 15 structured per-problem "interviewer briefs" (solution tree, 4-rung hint ladder, follow-ups, edge cases) driving interviewer behavior via the system prompt; layered cost protection with per-IP rate limits, session caps, and ephemeral keys.

**all-in-one-URL — Short URLs, QR Codes & Barcodes** | Python, FastAPI, PostgreSQL, Redis, Docker, React, TypeScript, Tailwind CSS

[all-in-one-url.vercel.app](https://all-in-one-url.vercel.app) · 2024

- Built a FastAPI backend for short URLs, QR codes, and barcodes with per-resource analytics, Redis-backed caching and counter batching, and tiered SlowAPI rate limiting.
- Implemented JWT auth with bcrypt hashing and per-resource ownership (preserving anonymous resources); containerized with Docker Compose and deployed across Vercel, Render, Neon (Postgres), and Upstash (Redis).

**Roofing Site & AI Voice Booking Agent** | Astro, TypeScript, Tailwind CSS, ElevenLabs, Cal.com API, Vercel

[premier-roofing-co.vercel.app](https://premier-roofing-co.vercel.app) · 2025

- Shipped an Astro + Tailwind marketing site for a local roofing business with a rule-based chat assistant, multi-step quote estimator, project gallery, and SEO-optimized routing.
- Integrated an ElevenLabs voice agent with Cal.com API webhooks to answer roofing FAQs, triage emergency requests, and book real inspection appointments end-to-end without human handoff.

**Modern Web Dev Guide — Interactive 2026 Web Dev Course** | Docusaurus, TypeScript, React, MDX, Mermaid

[tonyx1998.github.io/modern-web-dev-guide](https://tonyx1998.github.io/modern-web-dev-guide) · 2026

- Built an interactive quiz engine on top of the guide — a ~700-question bank with per-session seeded sampling, mandatory checkpoints, and progressive sidebar locking that gates each chapter behind prereq quiz passes.
- Authored 11 chapters spanning web foundations, project lifecycle, the modern tech stack, AI integration, and career paths; deployed via Docusaurus + MDX + Mermaid on GitHub Pages.

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## EDUCATION

**University of Maryland, College Park — College Park, MD**

Bachelor of Science, Computer Science · December 2022

**Relevant coursework:** Data Structures, Artificial Intelligence, Data Science, Computer Systems, Web Development.